Noah Novšak
Povšetova ulica 44, 1000 Ljubljana
Computer and Information Science, MAG
63230470

Committee for Student Affairs
University of Ljubljana, Faculty of Computer and Information Science
Večna pot 113, 1000 Ljubljana

Masters Thesis Topic Proposal

Noah Novšak

I, Noah Novšak, student of the 2nd cycle study program at the Faculty of Computer and Information Science, submit the following thesis topic proposal to be considered by the Committee for Student Affairs.

Working title:
Slovene: **Programski pristop k domnevi $\mathbf{PPT^2}$**
English: **A Software Approach to the $\mathbf{PPT^2}$ Conjecture**

This topic was already approved last year: **YES**

I declare that the mentors listed below have approved the submission of the thesis topic proposal described in the remainder of this document.

I would like to write the thesis in English with the following reason: I am a student of the Data Science program.

I propose the following mentor:
doc. dr. Aljaž Zalar
Faculty of Computer and Information Science
aljaz.zalar@fri.uni-lj.si

I propose the following co-mentor:
prof. dr. Igor Klep
Faculty of Mathematics and Physics
igor.klep@fmf.uni-lj.si

Ljubljana, December 5, 2025.

Key-words

quantum communication, positive-partial-transpose squared conjecture (PPT^2), positive not completely positive maps, semidefinite programming, entanglement breaking, separable states, positive polynomials, sums-of-squares polynomials

Detailed thesis proposal

Problem & State of the Art The PPT^2 conjecture posits that the composition of any positive-partial-transpose (PPT) map with itself is entanglement breaking. This hypothesis is critical for quantum communication, particularly regarding the security and feasibility of quantum repeaters [1,2]. While the conjecture is proven for low dimensions ($n \leq 3$) [3,4] and specific classes such as Choi type maps [5] and Gaussian quantum channels [4], it is widely believed to fail in higher dimensions ($n \geq 4$). However, no general proof or counterexample exists. A MATLAB-based library exists for generating map candidates based on positive polynomials that are not sums-of-squares [6,7], but it is limited in scalability and performance due to the complexity of the underlying semidefinite programs (SDPs) and MATLAB's inefficiencies.

Expected Contributions / Technical Outcome This work will deliver a high-performance, open-source software library in Julia capable of generating random positive maps that are not completely positive (PnCP) in higher dimensions. The primary contribution is a rigorous computational stress-test of the PPT^2 conjecture: we aim to either isolate a numerical counterexample, thereby disproving the conjecture, or provide significant statistical evidence of its validity. Additionally, the project will expand the tooling landscape by implementing methods to generate maps that are strictly k-positive but not $(k+1)$-positive, providing new resources for analyzing the geometry of quantum channels.

Methodology & Validation We will re-architect the existing generation framework using Julia and the JuMP package to interface with high-performance solvers like MOSEK. The approach relies on polynomial sum-of-squares (SOS) relaxations to generate map candidates, followed by semidefinite programming (SDP) to verify separability criteria [8,9]: first, we will benchmark the new library against the legacy MATLAB implementation to quantify improvements in runtime, memory usage, and dimensional scalability. Second, we will deploy large-scale randomized sampling to probe the boundaries of the positive map cone, using SDP certificates to validate the entanglement-breaking properties of the composite maps.

References

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[4] M. Christandl, A. Müller-Hermes, and M. M. Wolf, When Do Composed Maps Become Entanglement Breaking?, Ann. Henri Poincaré **20**, (2019).

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